POLITECNICO MILANO 1863

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ATLANTIC SALMON

BIOINSPIRED ROBOT

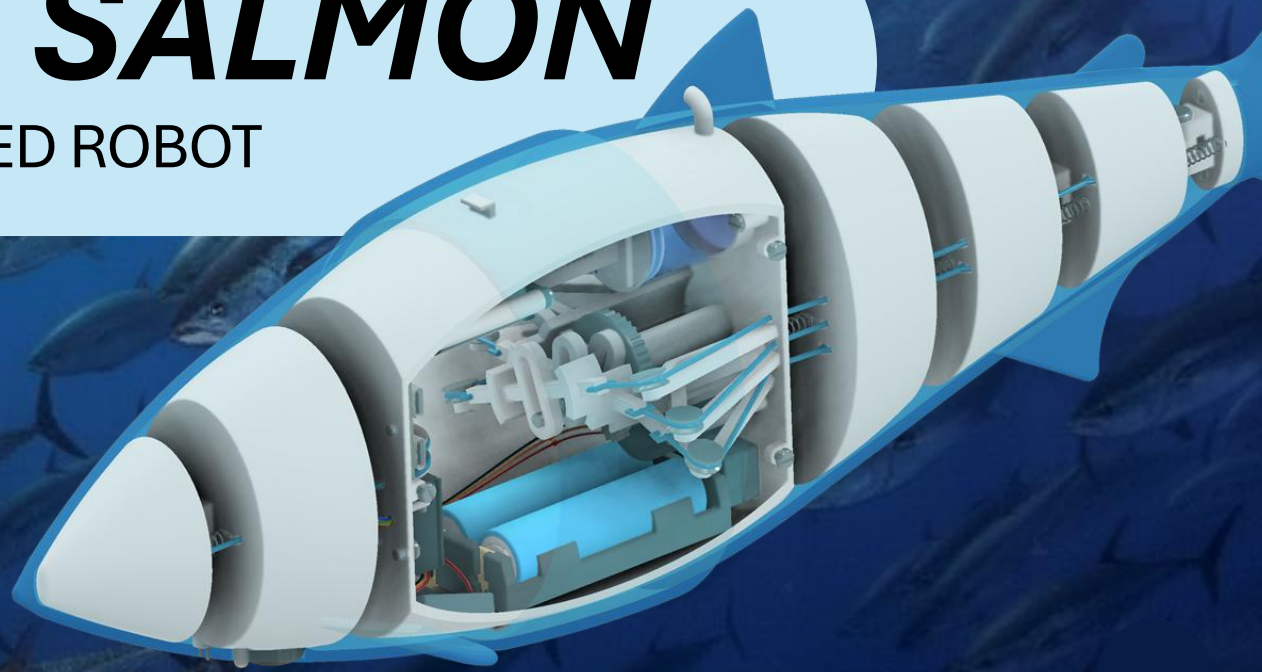


TABLE of CONTENTS

1 ***Introduction***
- *Morphology*

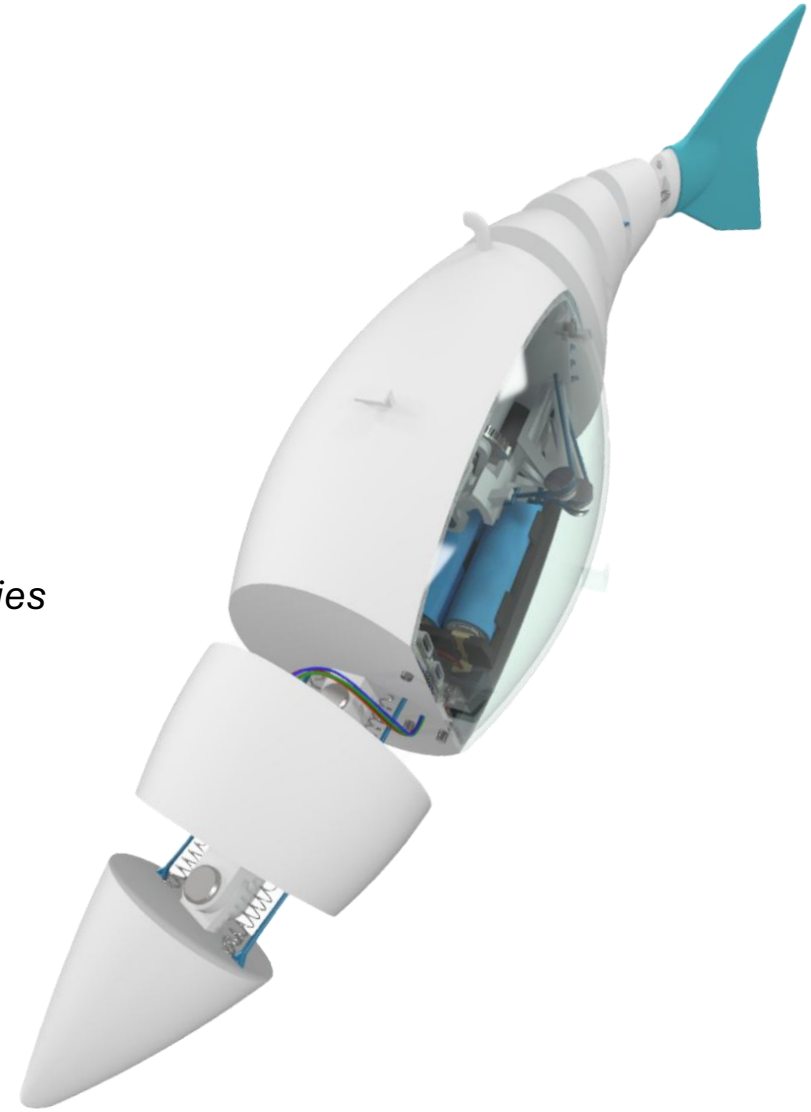
2 ***Locomotion***
- *Robot design overview*
- *Performance of the fish*
- *Stiffness computation*

3 ***Robot design***
- *Actuation*
- *Waterproof assurance*
- *Electronics & Sensors*
- *Assembly*

4 ***Application***

5 ***Simulation results***
- *Simulated motion*
- *Kinematics*
- *Displacement & Velocities*
- *Forces*
- *Torques*
- *Thrust*
- *Cost of Transport*

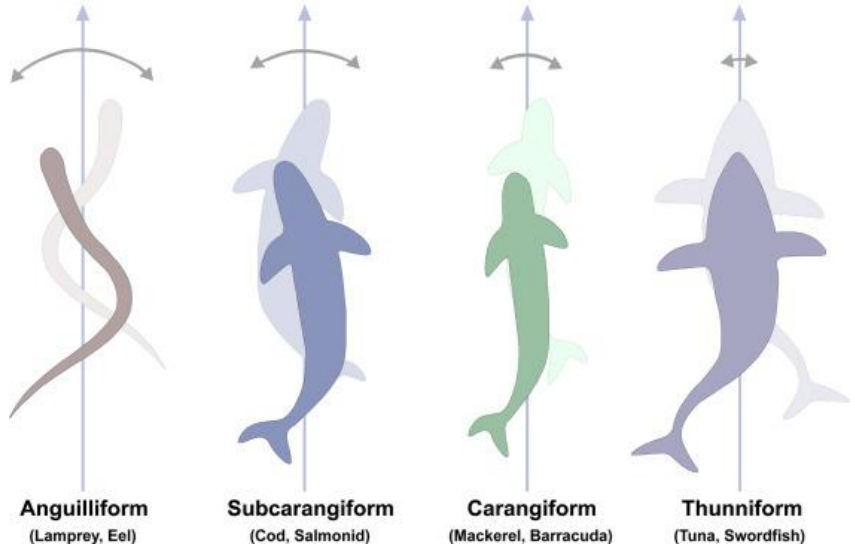
6 ***References***



1. INTRODUCTION

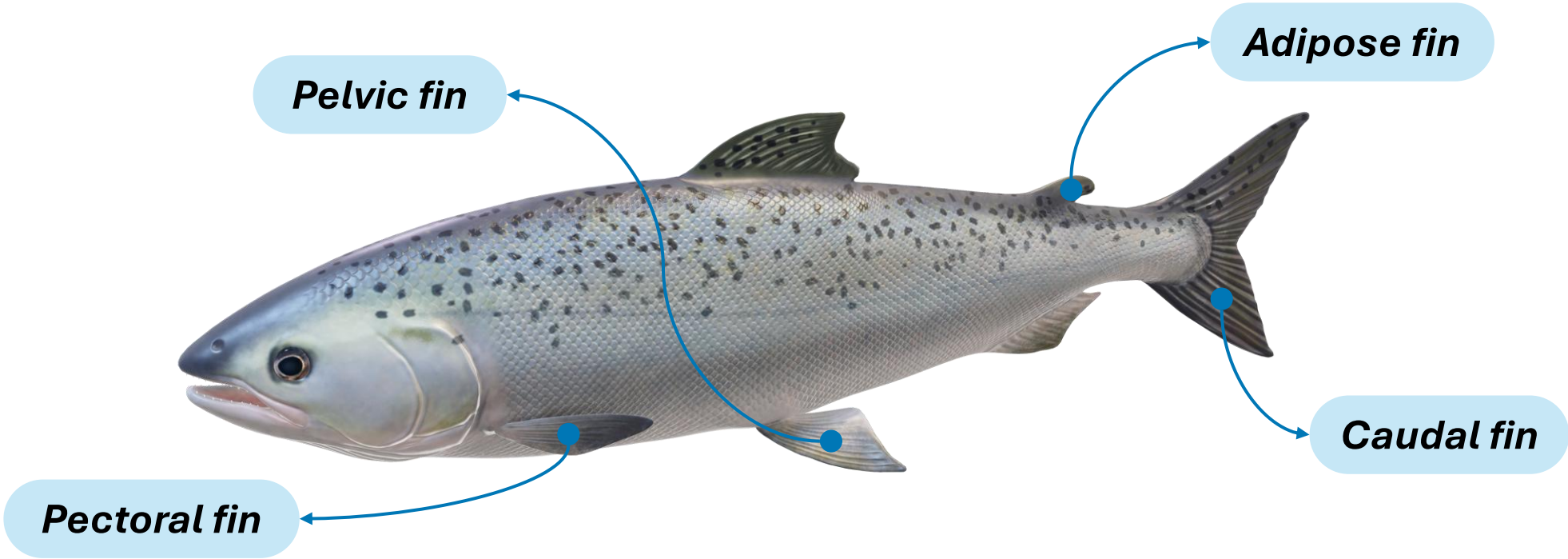
→ The Atlantic salmon (‘*Salmo Salar*’) belongs to the family of the Salmonidae and can reach dimensions of 1 meter long.

→ It’s type of locomotion is categorized in the field of BCF fishes as a ***subcarangiform***.



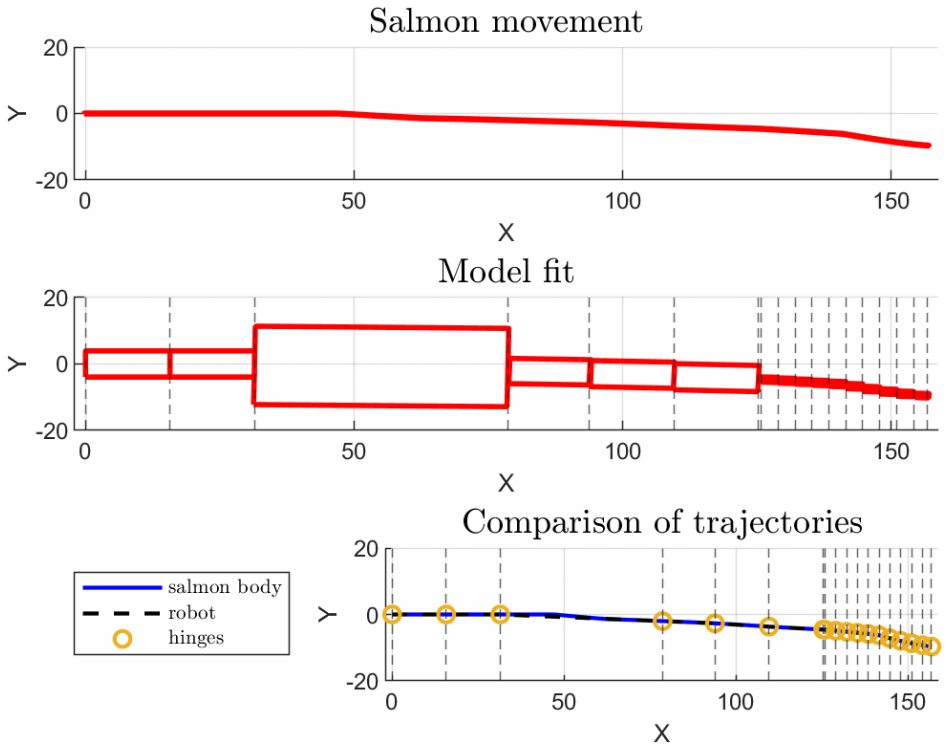
1a. Morphology

→ The body of the Atlantic salmon is elongated and fusiform to minimize the hydrodynamic drag during movement.



2. LOCOMOTION

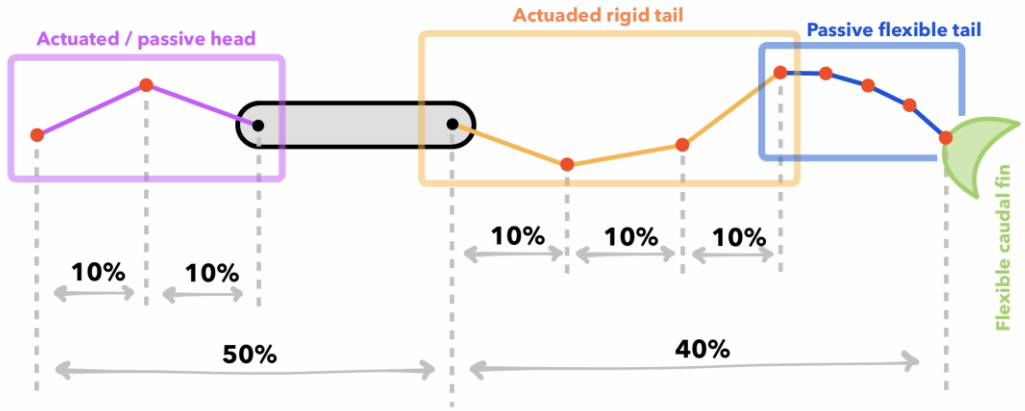
→ The Salmo salar is categorized as a subcarangiform so the main thrust is generated moving 1/3 - 1/2 of its body in a sinusoidal way.



2a. Robot design overview

→ The robotic salmon is structured in the following way:

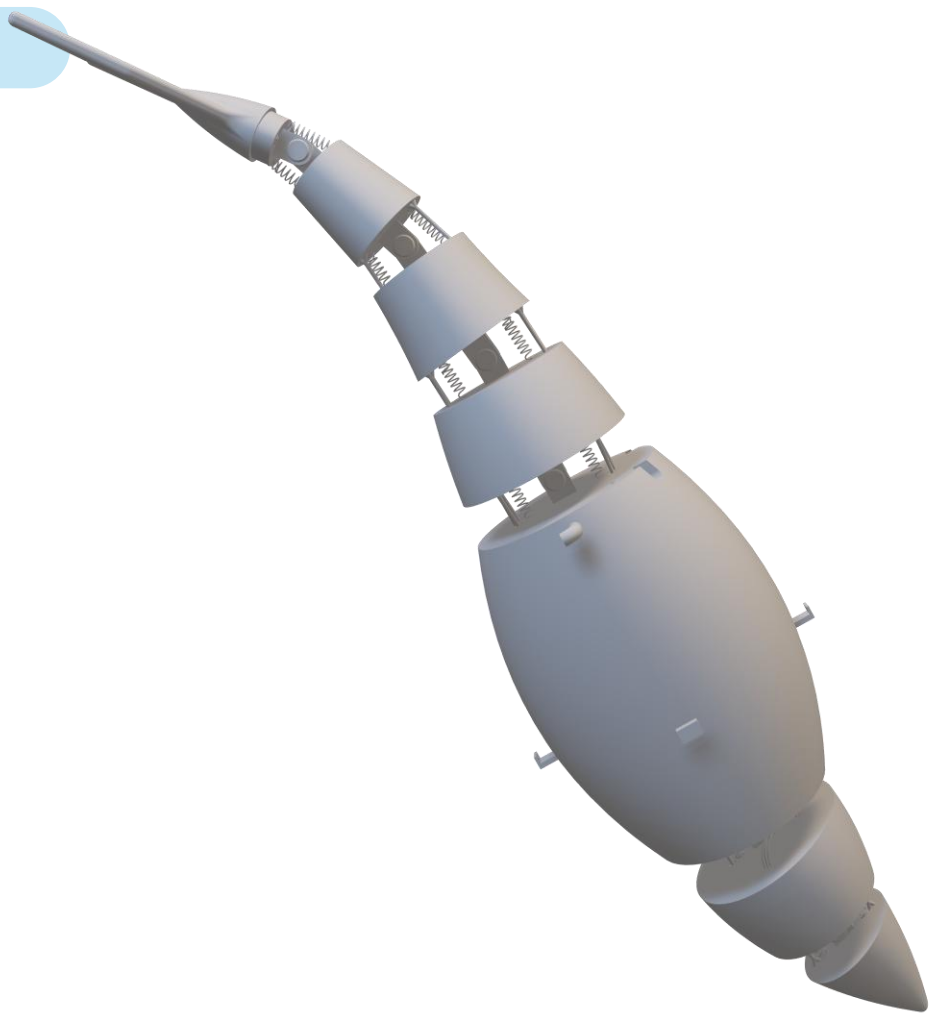
- ***Head***: consists of two links
- ***Mid-body***: includes the *mechanical transmission* for the head and tail, the *power supply unit* and the *controller unit*
- ***Rigid tail***: consists of three links
- ***Flexible tail***: made of a soft-silicone material



2a. Robot design overview

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→ To mimic the locomotion some characteristics must be maintained:

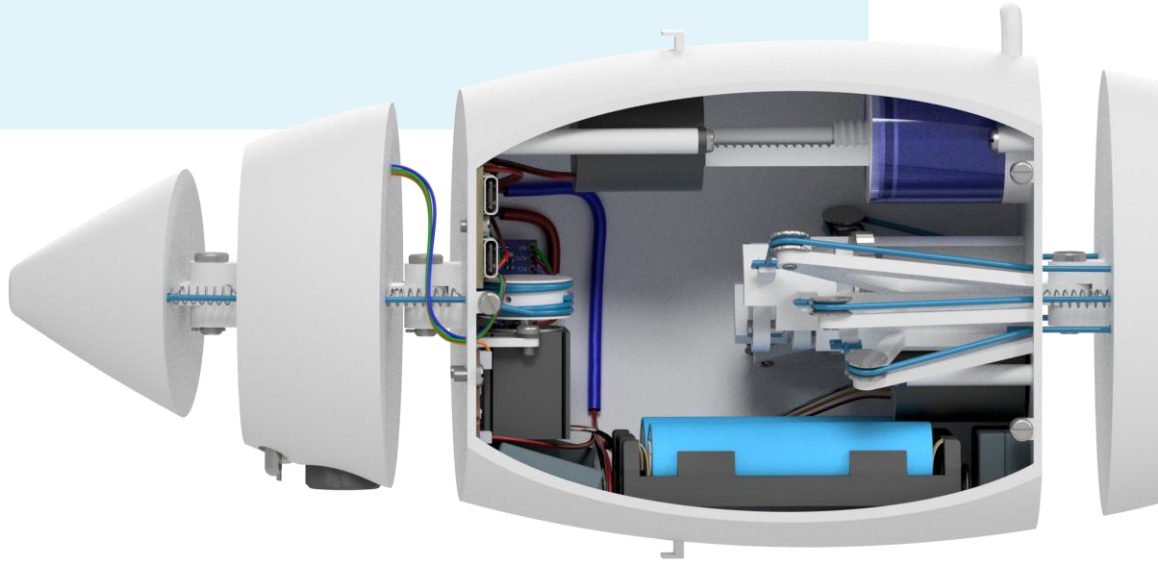
Slender body

Goal
→ minimize the **drag** improving the overall **efficiency**

How to implement
→ follow the shape of a real Atlantic salmon
→ Cover the overall robot with a **silicone skin**

High manoeuvrability

How to implement
→ two **actuated links** in the head driven by an independent motor



2b. Performance of the fish

→ An important observation that can be made is how the fish adapts certain parameters during a change of velocity to maintain a *high efficiency*

Velocity rise [1],[2]

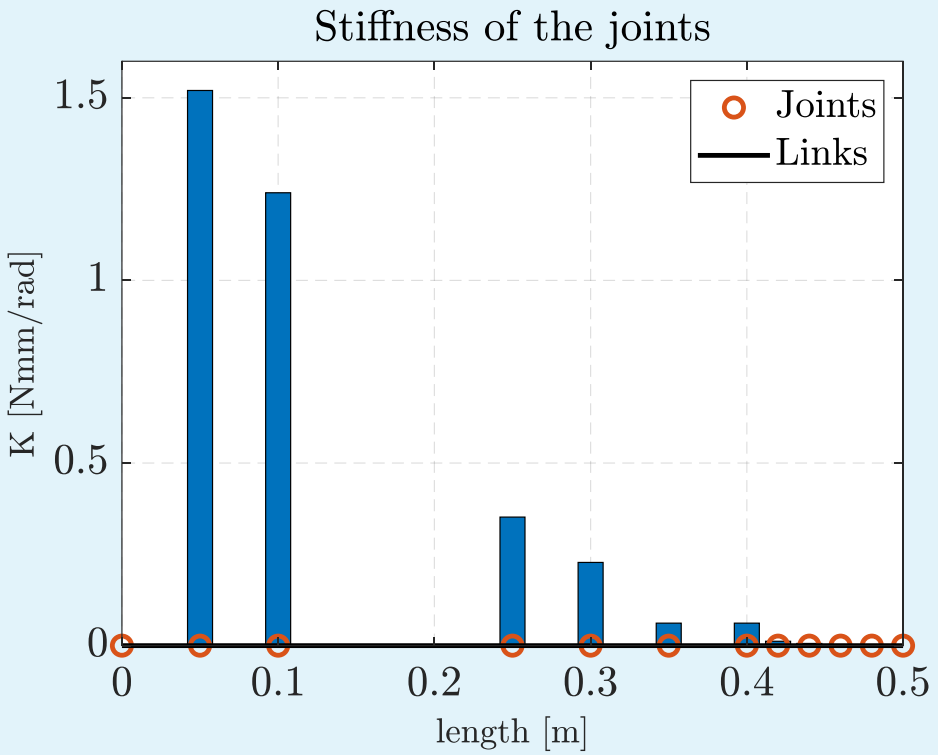
- The swimming ***frequency*** *increases*
- The ***amplitude*** of the travelling wave *increases*
- The fish becomes less flexible, so the ***stiffness*** *increases*

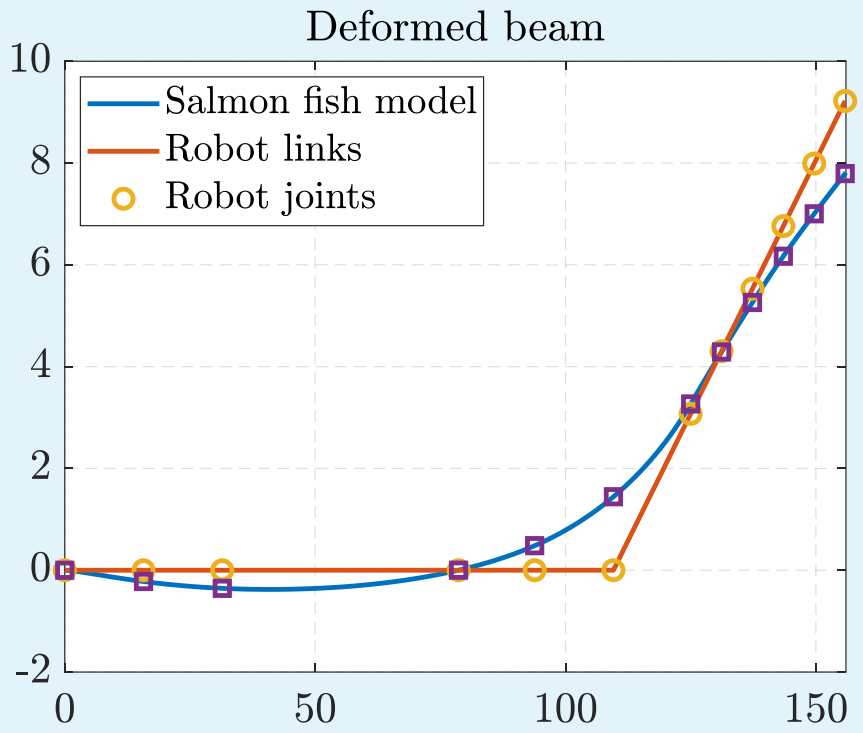
2c. Stiffness computation

→ An important parameter to derive are the ***stiffness*** that should be applied to each joint, the following steps are done:

1. Take the useful parameters from a test performed in a paper [1]
2. Recreation of the fish and of the robot in a simulation environment
3. Application of a momentum to the modelled fish
4. Do the same with the modelled robot adapting the stiffness to have similar results

→ The values extracted form the previous method are the following:





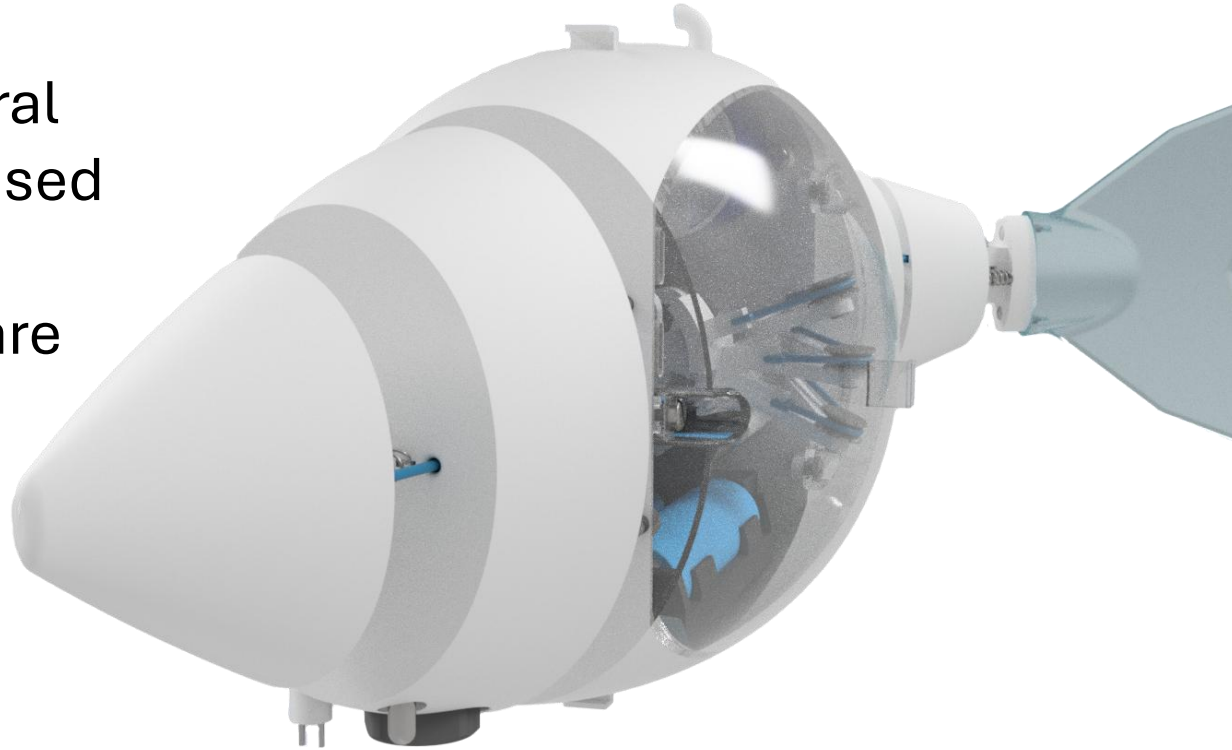
3. ROBOT DESIGN



3. ROBOT DESIGN

→ The robot consists of ***6 rigid links***, with one of them that contains the electronics needed.

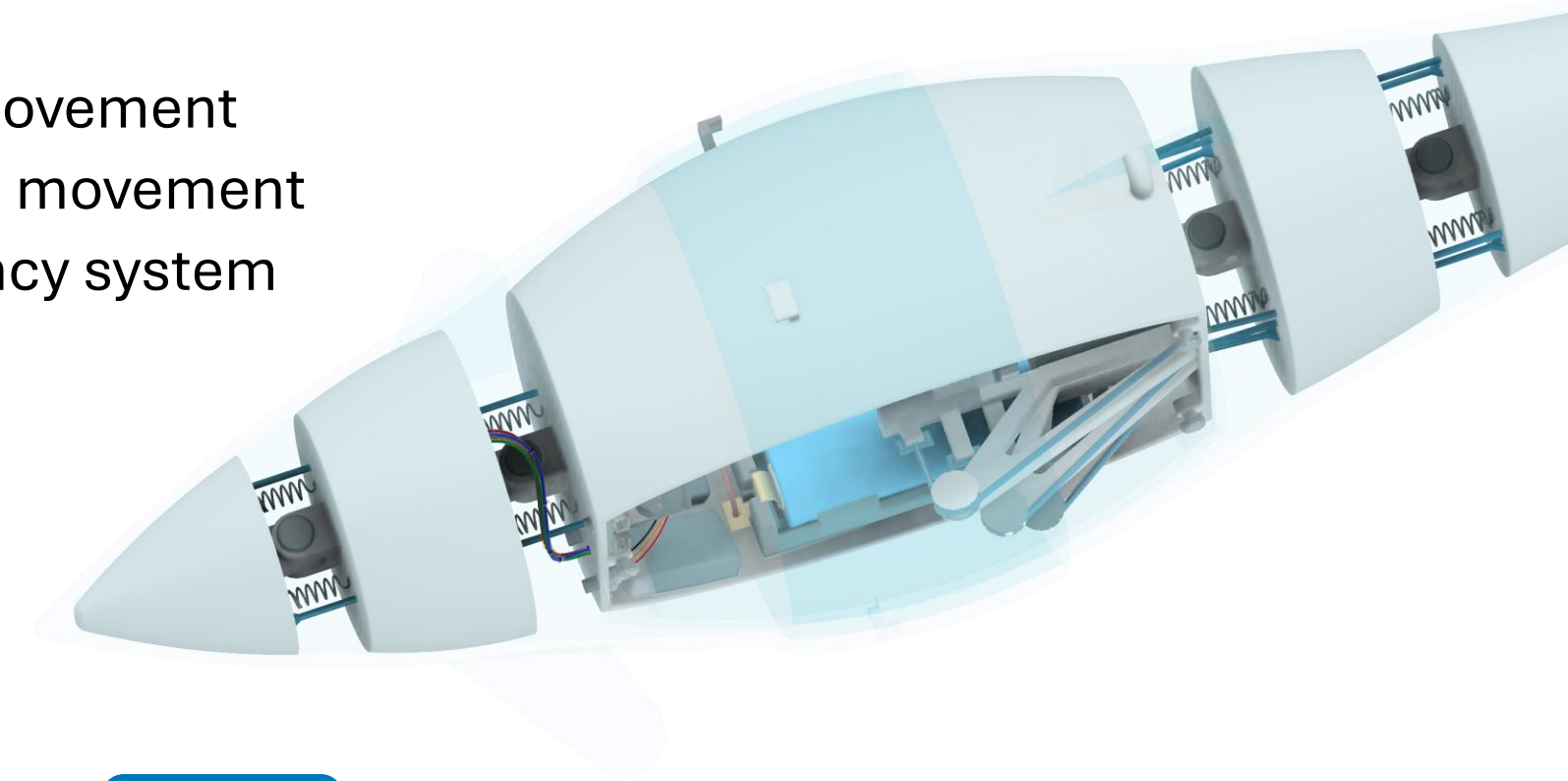
→ The main material used for the structural parts is ***ABS plastic***, while ***silicone*** is used to create the soft skin that completely encloses the robot. ***Dyneema strings*** are used to reduce friction and reduce backlash.



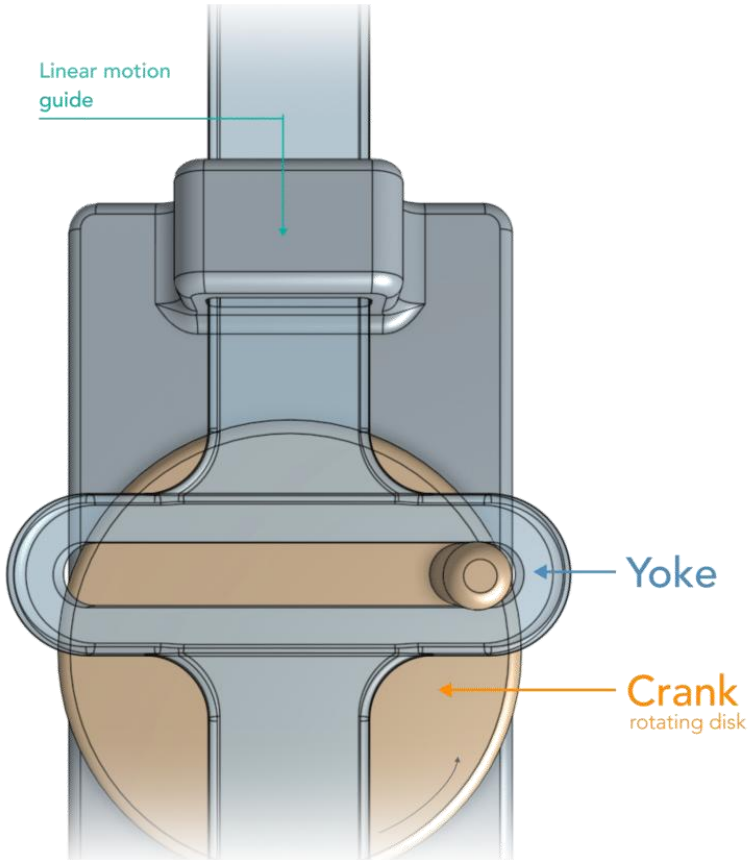
3a. Actuation

→ The actuation strategy is divided into three separated sections dedicated to the motion of a specific part with its own motor:

1. ***Main motor*** → Tail movement
2. ***Front motor*** → Head movement
3. ***Top motor*** → Buoyancy system



1 ***Main motor*** → *Tail movement*



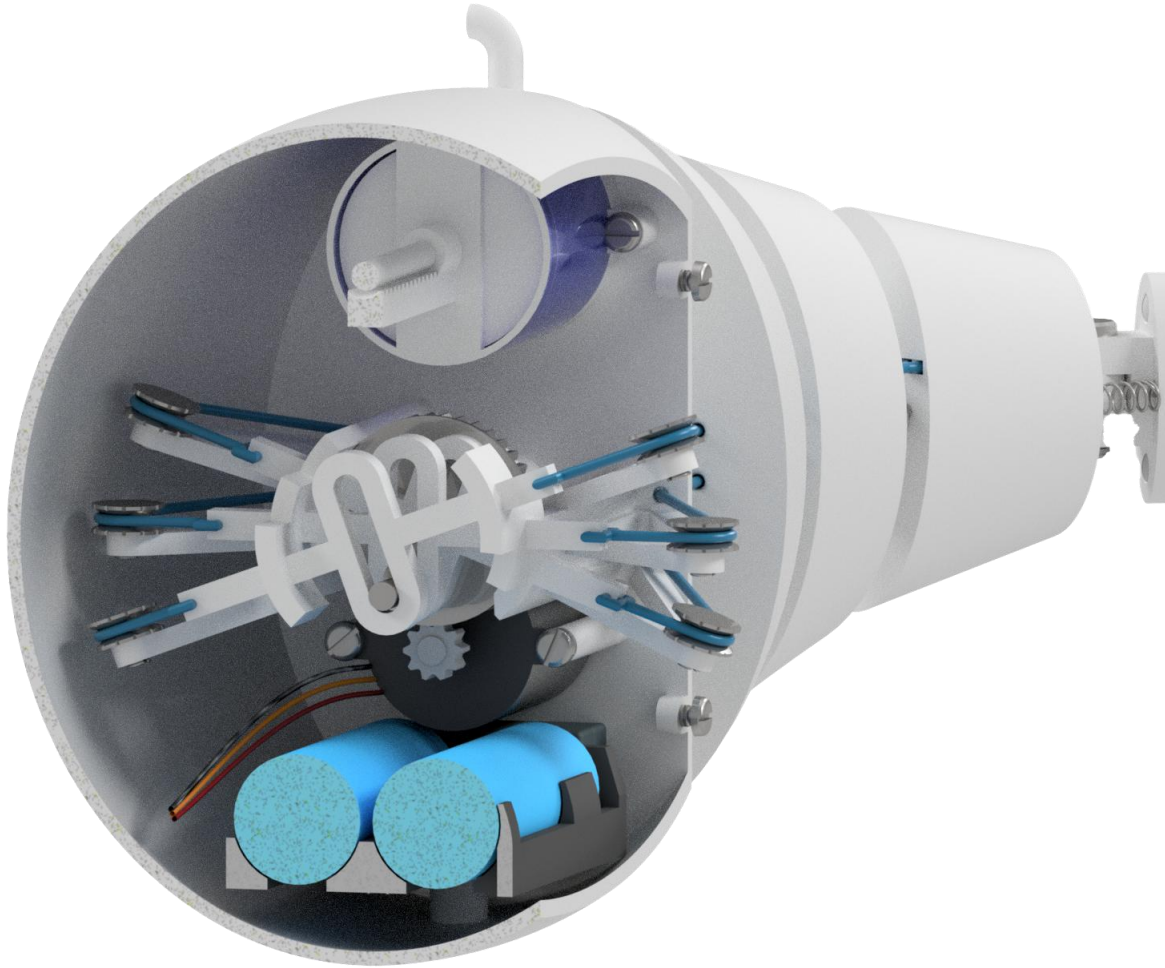
– The motor actuates three dependent ***Scotch Yoke mechanisms*** to move the active links of the tail to create a *travelling wave*.

1 ***Main motor*** → *Tail movement*

- A brushless motor of the series ***Skywalker 2312SL-1250KV*** actuates the motion of three Scotch Yoke mechanisms to recreate the travelling wave.

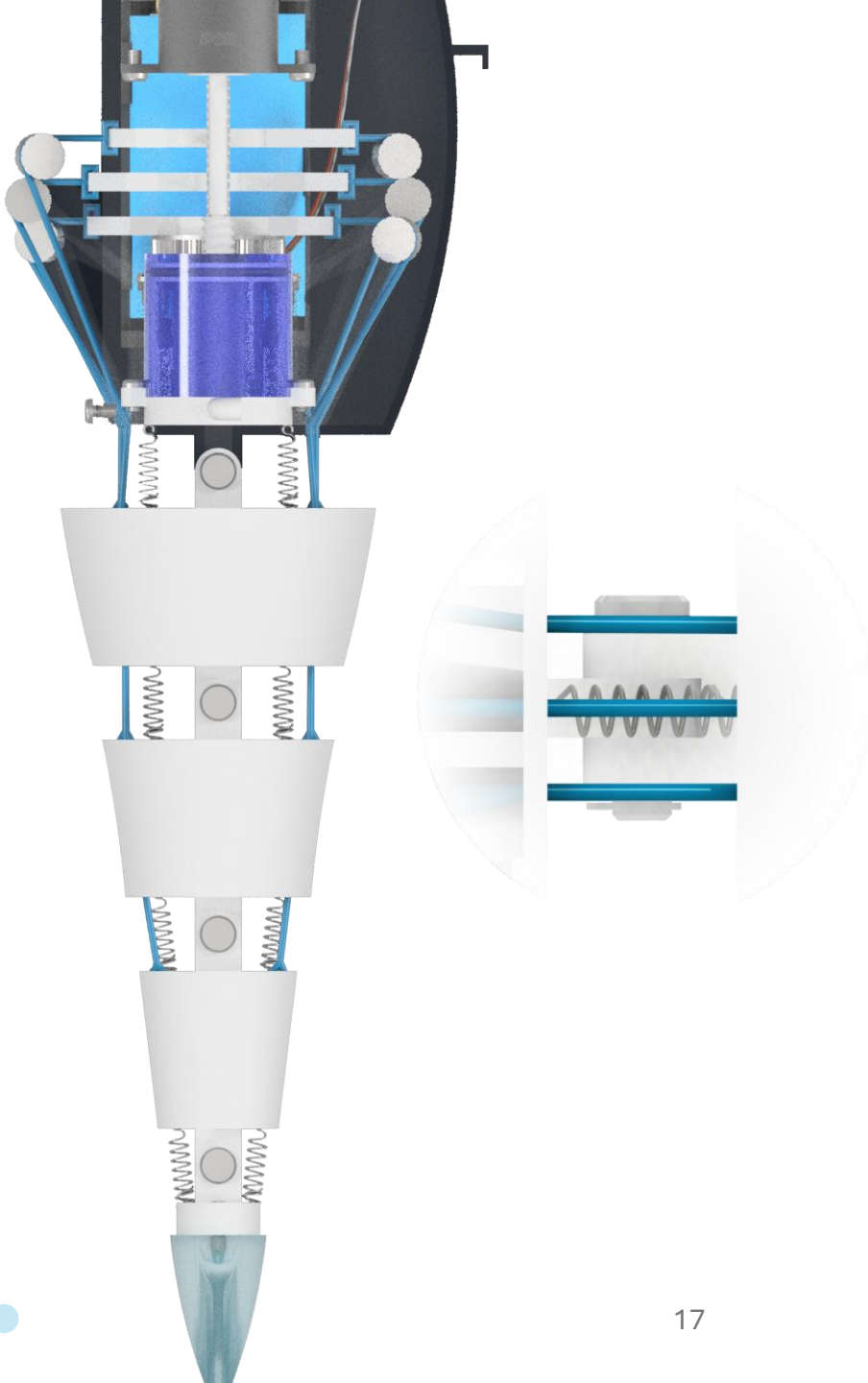
- The mechanism connects each link to a pair of cables and allows to have a ***phase shift*** between each of them.

$$\varphi = 20°$$



1 ***Main motor*** → *Tail movement*

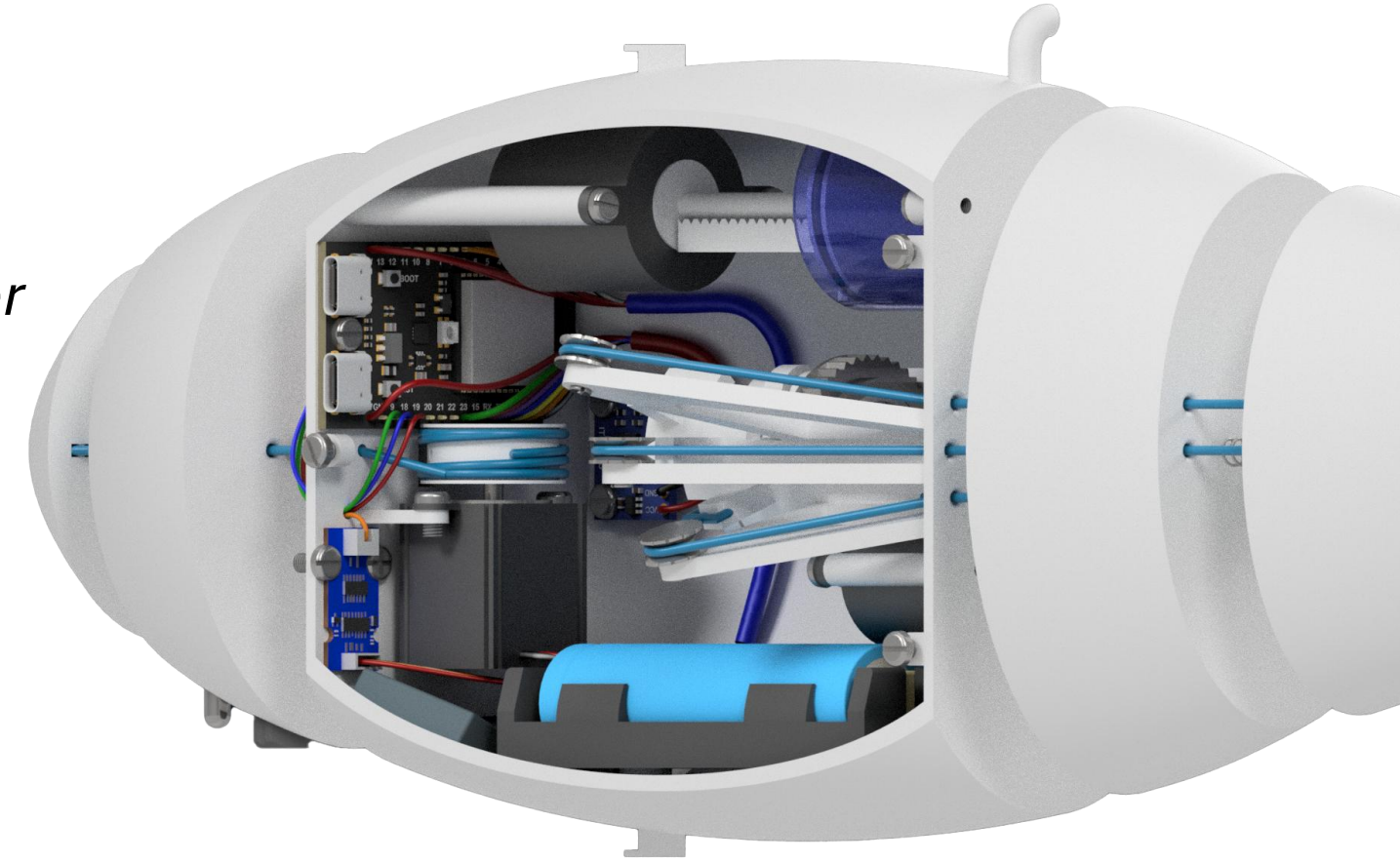
– The ***antagonistic cables*** are needed to ensure the correct movement. Each link is connected to the previous one using a *pivot* and a *spring* to replicate the fish stiffness.



2 ***Front motor*** → *Head movement*

- It actuates the two links of the head, as a rudder, to *steer* and gain more *agility* when necessary.

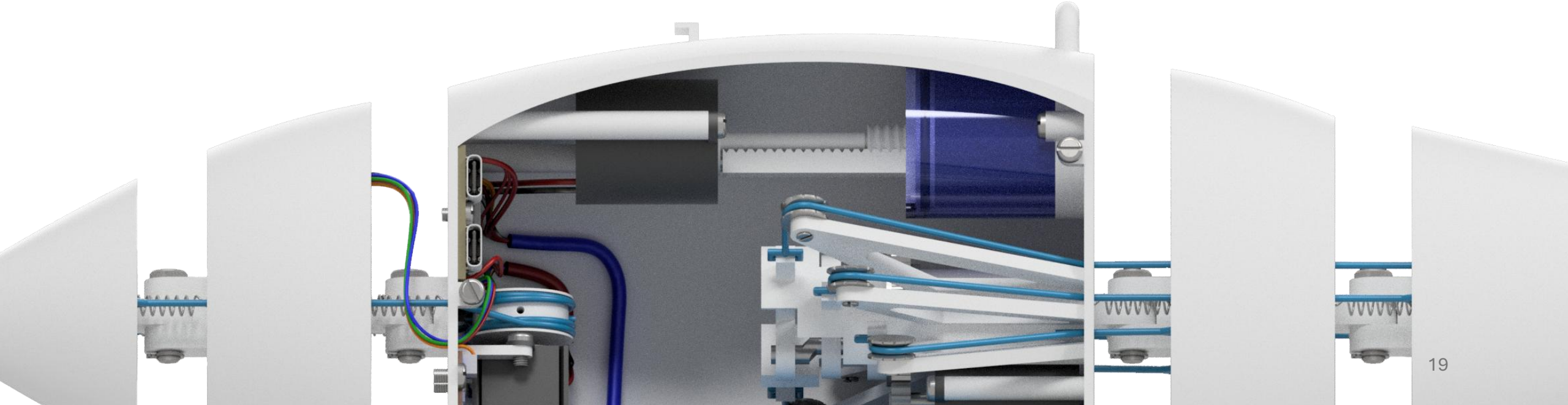
- It's a ***servo motor*** that through a pulley moves the rope to command the links



3 ***Top motor*** → *Buoyancy system*

- The motor drives a ***worm gear and rack*** mechanism to *fill* or *drain* a reservoir that allows the robot respectively to ***submerge*** or ***emerge***.

$$V_{max-d/a} = 0{,}23\ m/s$$



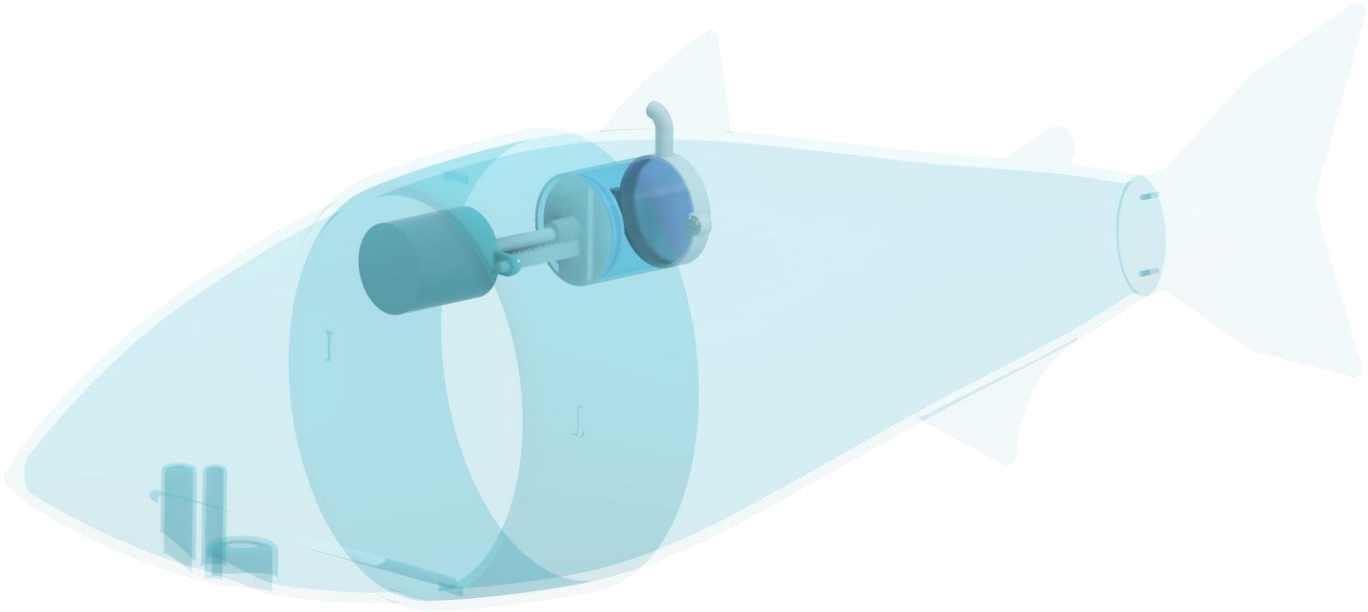
3 ***Top motor*** → *Buoyancy system*



– The robot is also been weighted to recreate a ***neutral buoyancy*** and lower the barycentre by using adhesive lead weights.

3b. Waterproof assurance

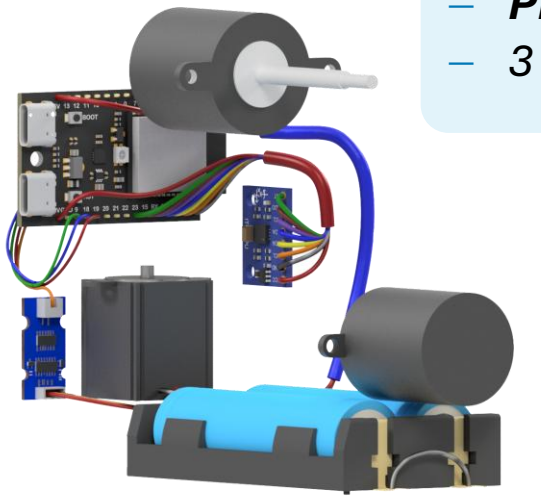
→ To ensure a complete enclosure of the mechanical and electrical parts of the robot an ***external silicone skin*** is developed. It has been split in two parts to simplify the fitting of it. Some ***staple shaped brackets*** maintain it tight and in position.



3c. Electronics & Sensors



→ For the control and analysis of information some auxiliary electronic components are needed.

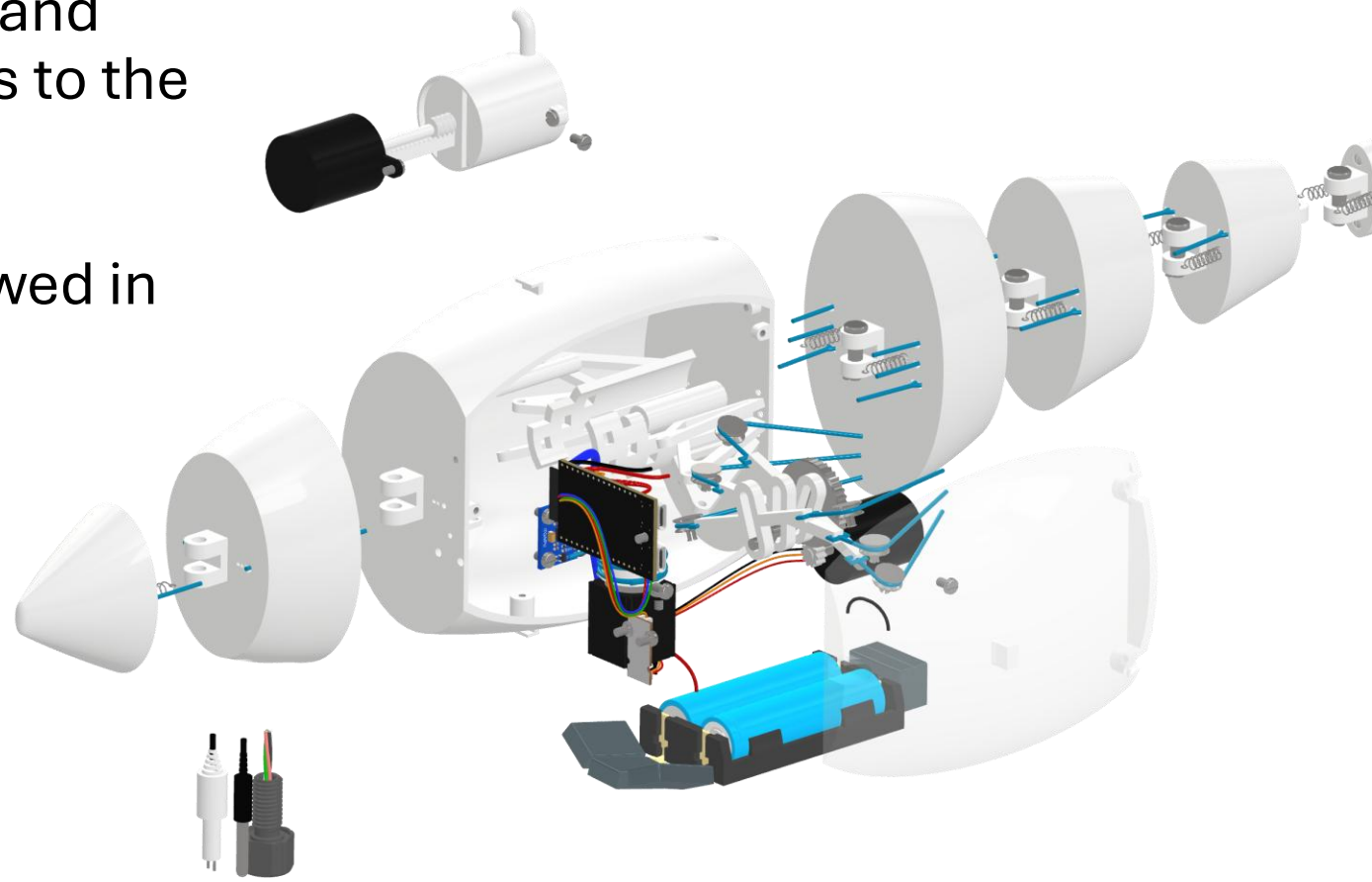


- *Waterproof* ***temperature sensor*** - S18B20
- ***Groove TDS*** (Total dissolved solids)
- ***Pressure sensor*** - BAR30-ASM-R2
- *3 DOFs* ***accelerometer and gyroscope*** - MPU-6050

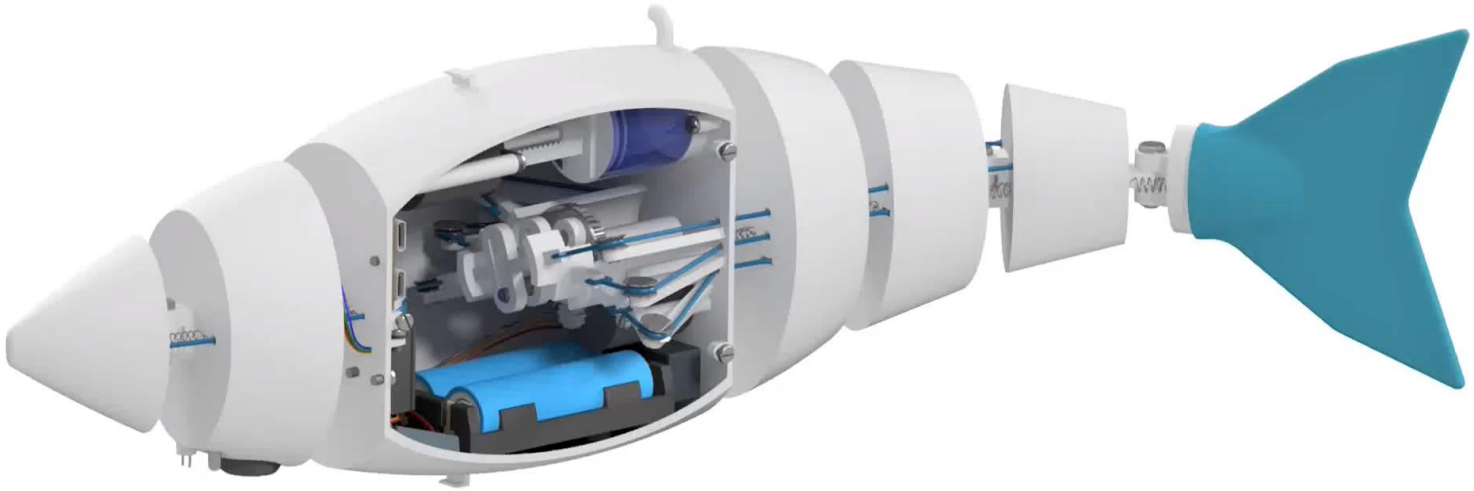
- ***ESP32 controller*** for compatibility
- *Two* ***18650 3.7 V batteries*** wired in series

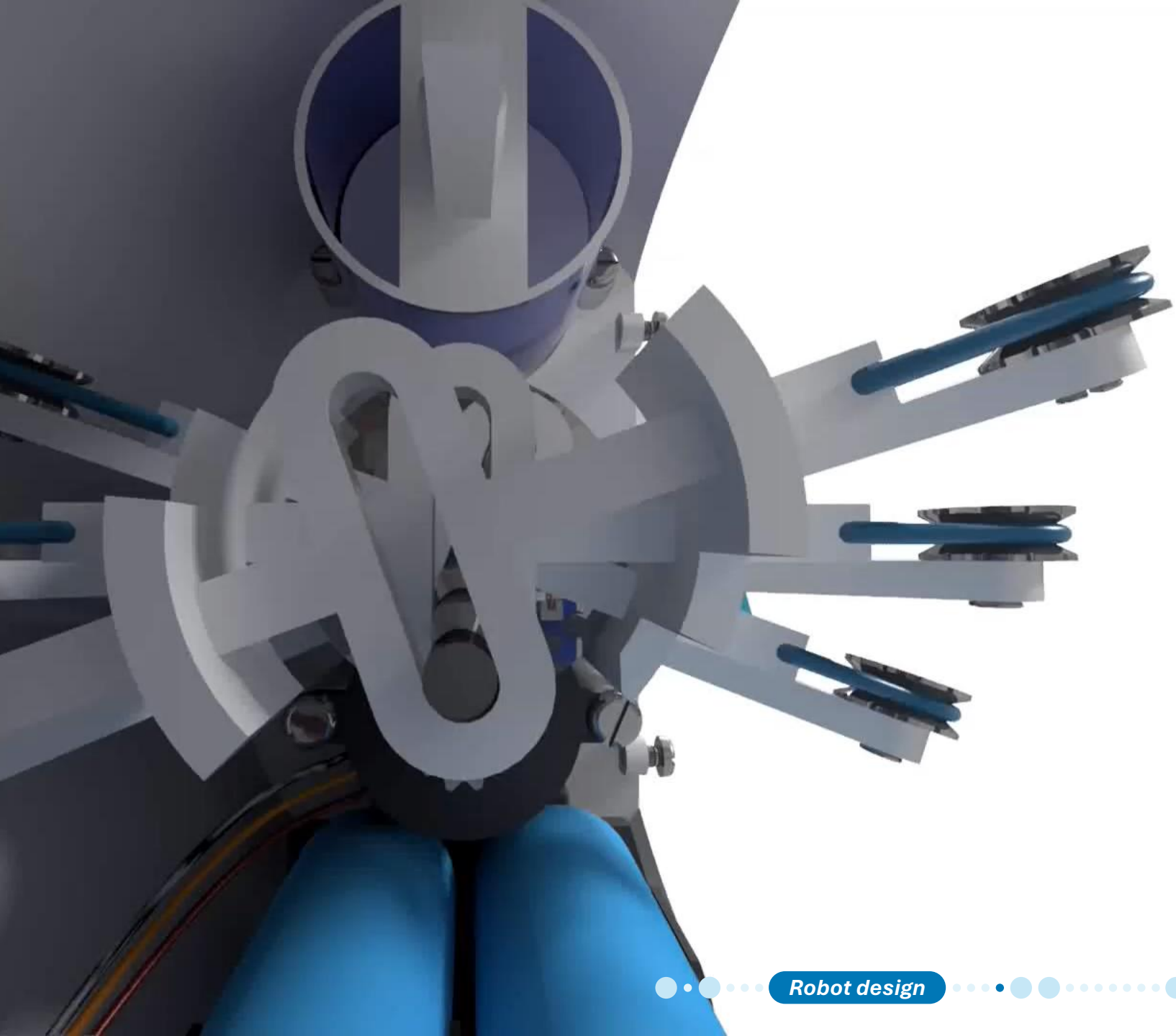
3d. Assembly

→ To assemble the links ***steel pins*** and ***Seeger rings*** are used. The access to the inside of the robot is allowed by a ***removable cover***. All the other components are ***press-fit*** or screwed in place with ***M3 screws***.



→ The animation shows the ***movement of the tail*** and of the ***head*** that can be used as a rudder or to improve the agility of the robot.

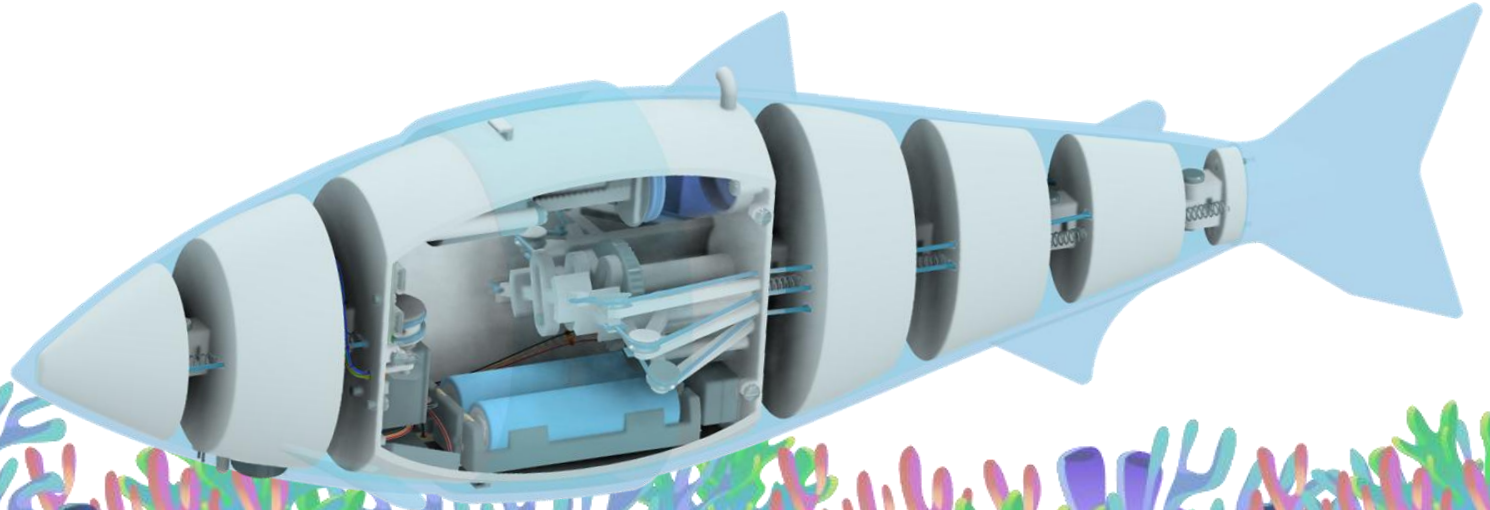




→ The motion of the three ***Scotch Yokes*** can be valued from the animation of the 3d model.

4. APPLICATIONS

→ Some possible uses of this type of bioinspired robot could be as a ***marine monitoring system*** of coral reefs, for example for the *water quality*, to detect *pollution* or to do some inspections on the coral health.



5. SIMULATION RESULTS

→ To perform the dynamics computation the ***Discrete Vortex Method*** is used obtaining in output the forces generated by the movement.

- *Mass of the fish* → $m_{tot} = 1{,}2\ kg$
- *Length of the links* → $L_{tot} = 0{,}5\ m$
- *Frequency* → $2\ Hz$
- *Stiffness*
- *Amplitude of the movement*

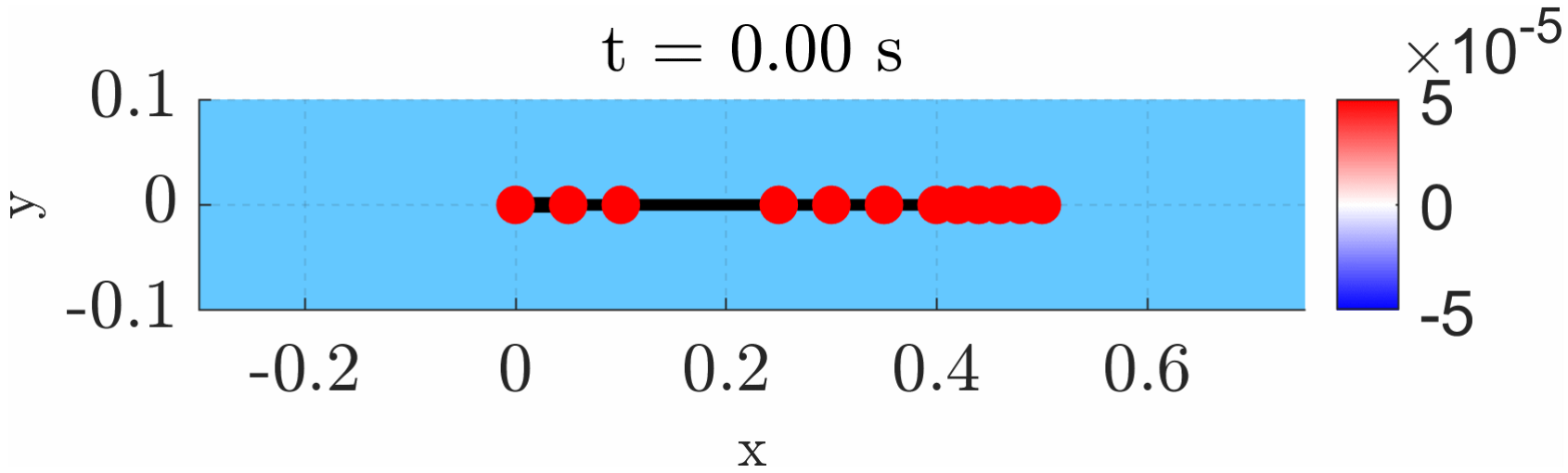


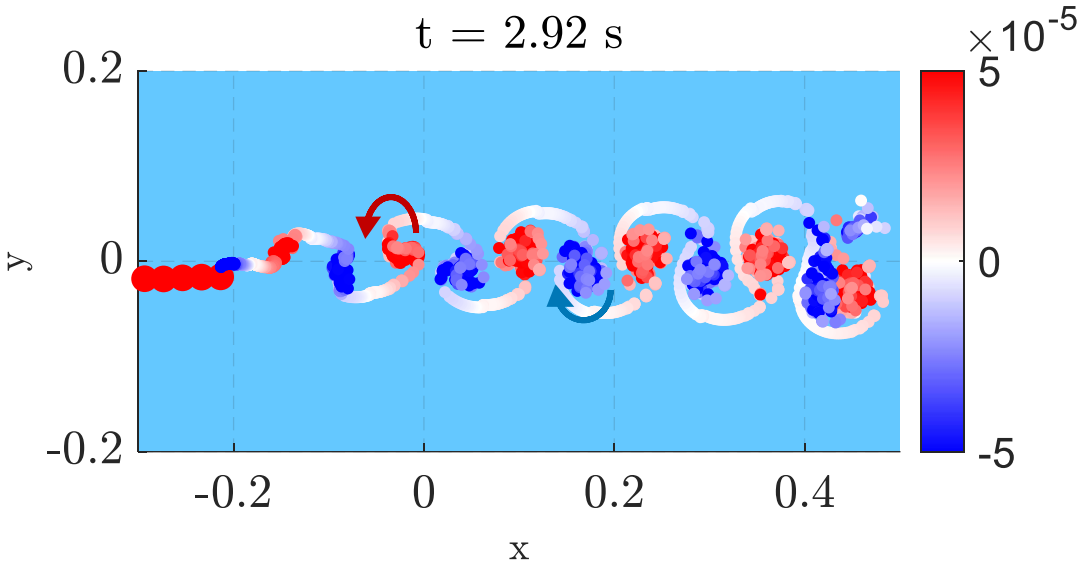
The parameters are then verified through a system with ***Reynolds*** and ***Strouhal*** number

$$\begin{cases} Re = \dfrac{UL_{tot}}{\nu} \cong [10^5 - 10^6] \\ St = \dfrac{2fAmp}{U} \cong [0{,}2 - 0{,}4] \end{cases}$$

5a. Simulated motion

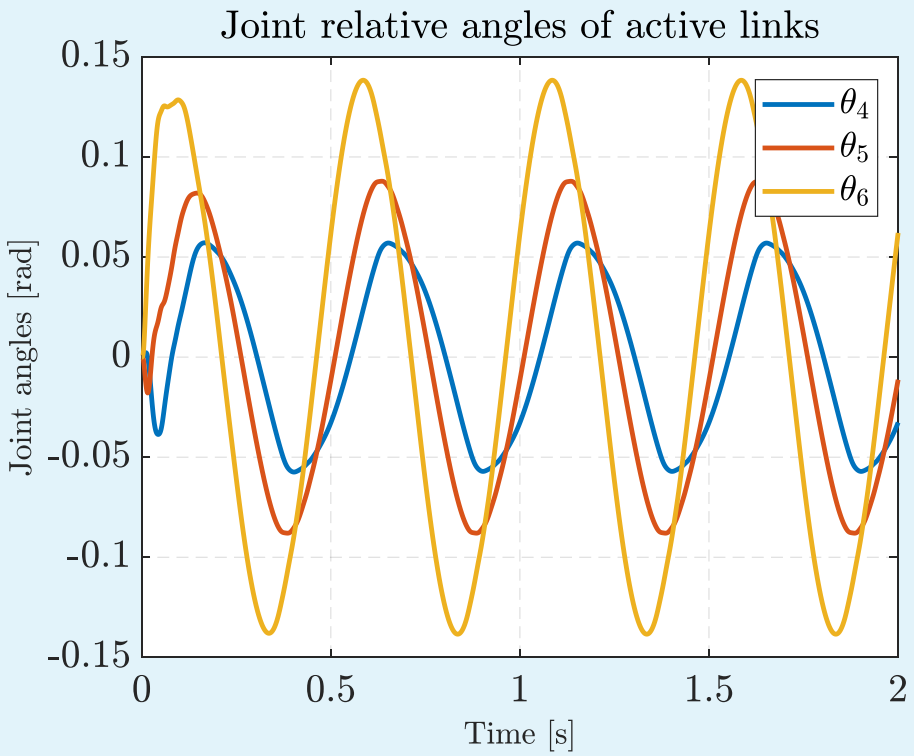
→ An output of the method are the ***vortices*** shed in the wake and the ***kinematic angles*** of the links, so a visual simulation of the motion can be plotted highlighting some useful information.





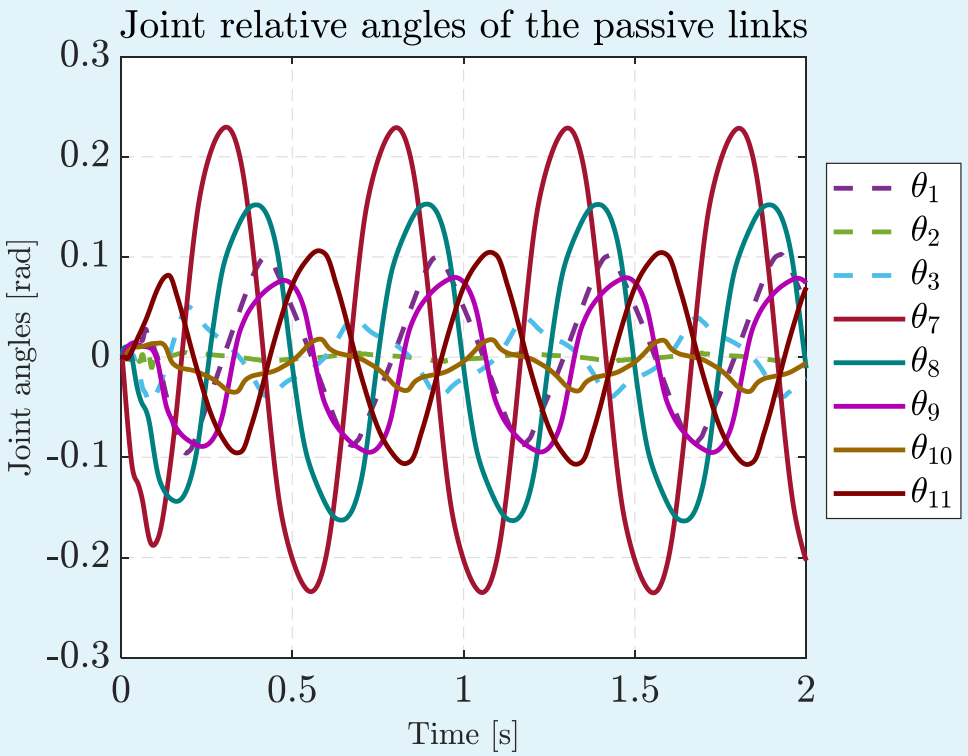
– The ***reverse Karman Street*** is maintained ensuring the thrust needed. The **red** vortices rotate counterclockwise, while the **blue** one with a clockwise direction providing a ***velocity surplus***.

5b. Kinematics



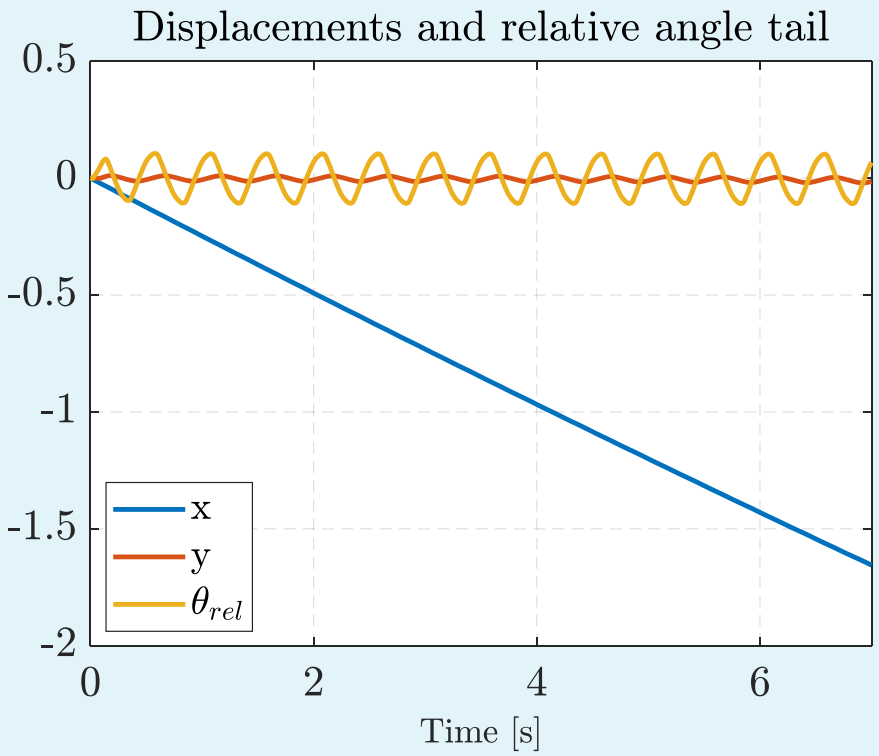
→ The movement of the active links is performed giving the ***sinusoidal law*** to each joint in terms of ***relative angles***. From the output of the simulation is possible to check if the system works correctly.

5b. Kinematics



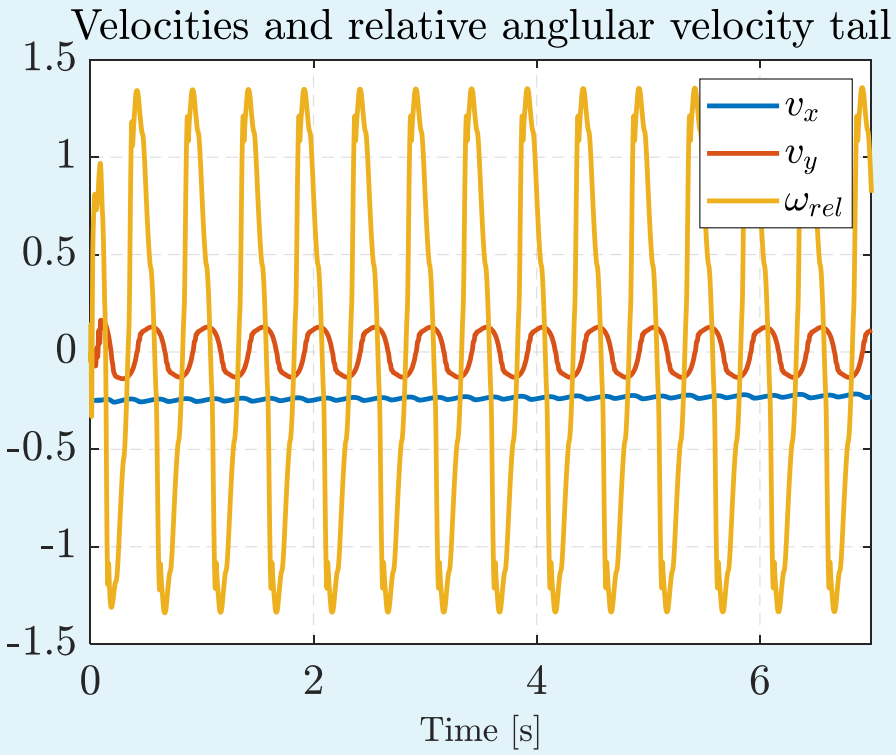
→ Looking to the ***passive links*** it's possible to notice that the amplitude of the tail related links decreases going towards the caudal fin, sign of a correct behaviour.

5c. Displacement & Velocities



→ The decomposed movement along the ***x*** and ***y direction*** can be visualized highlighting a *linear movement* in the direction of motion and a small *oscillating motion* in the vertical plane.

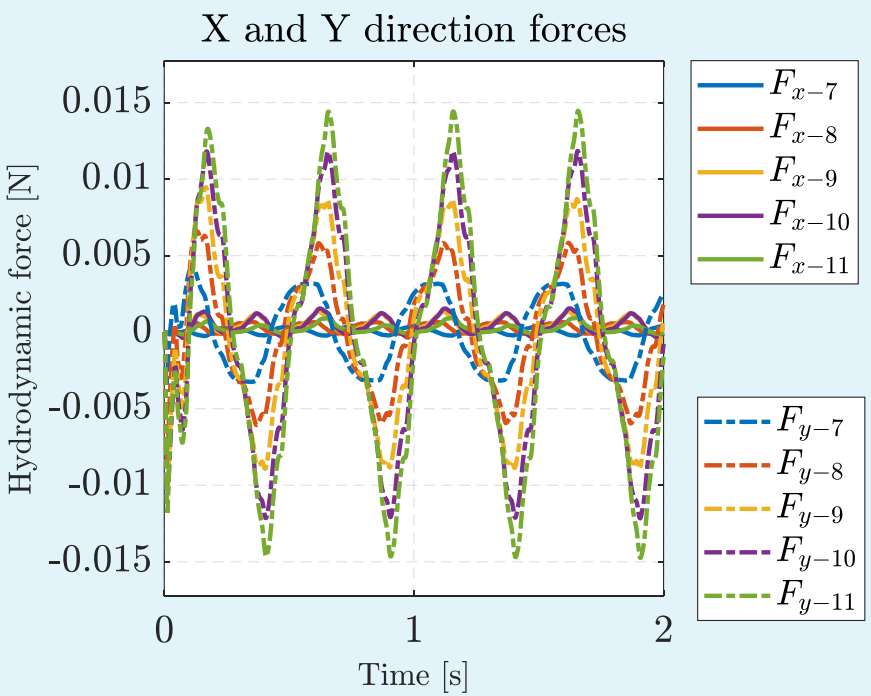
5c. Displacement & Velocities



→ From the ***velocities*** point of view the behaviour is similar showing a constant swimming velocity in the *x direction*

5d. Forces

→ The forces of the links considered to produce the thrust along the ***x*** and ***y direction*** can be plotted showing an oscillatory trend. The peak value of the force increases towards the ***caudal fin*** where the maximum is showed.

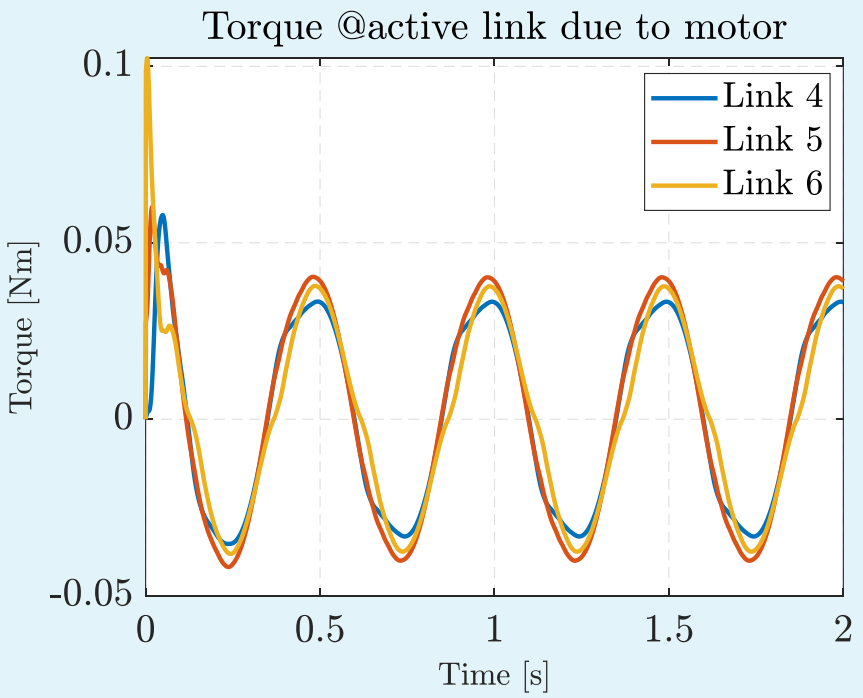


5e. Torques

→ The torques of the active links of the tail show a ***stable behaviour***. The ***phase shift*** is noticeable, and the torques values are consistent with the robot characteristics.

→ Knowing the required torque and the kinematics, the necessary power can be computed

$$P_{nom} \cong 0{,}6\ W$$

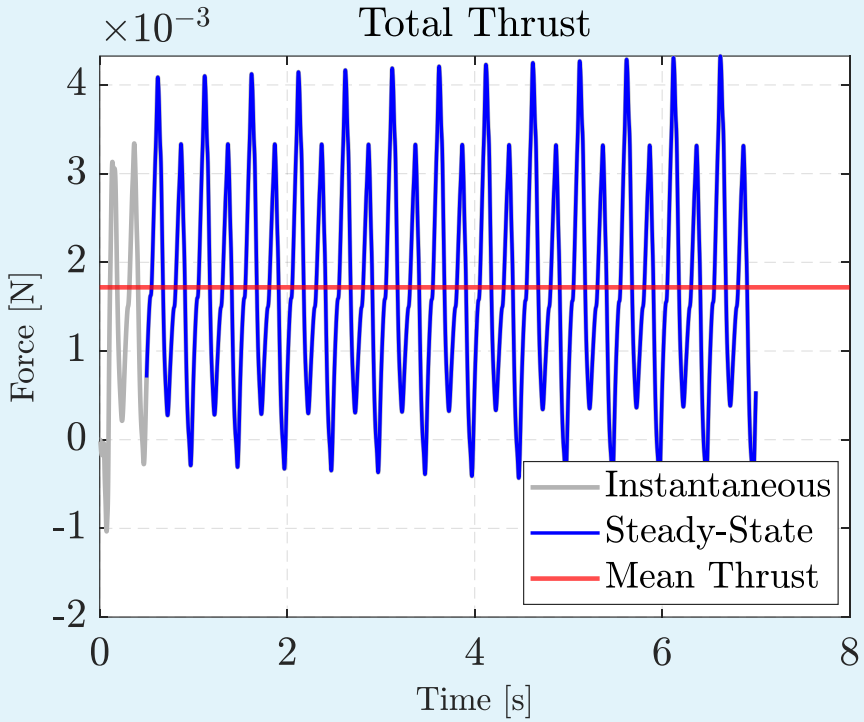


5f. Thrust

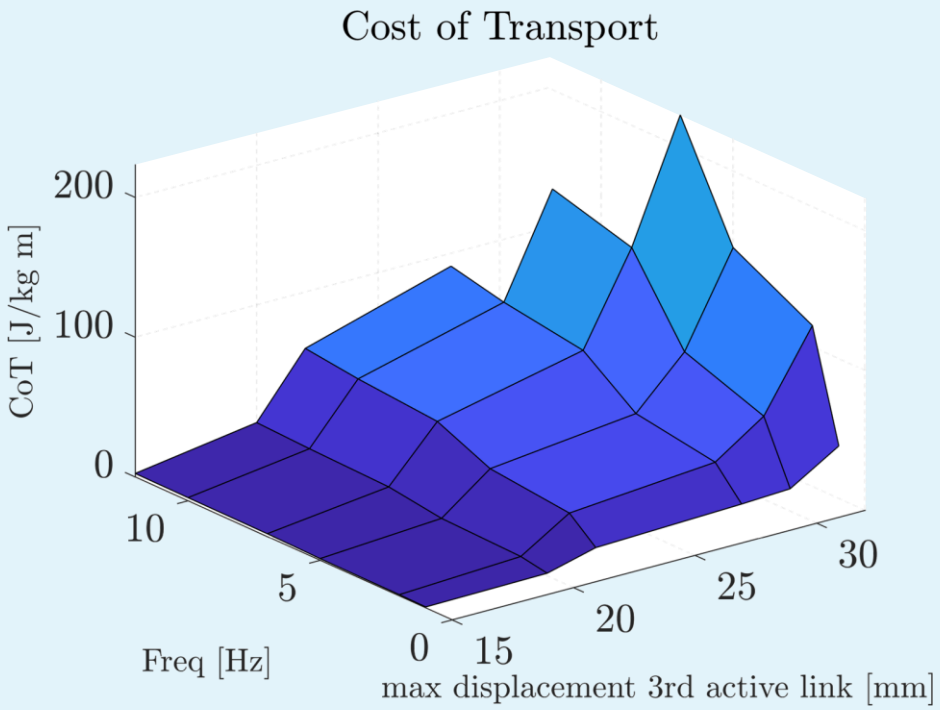
→ The total ***instantaneous thrust*** is the summation of the forces in the *x direction*, to compute the mean value the transient has been excluded.

$$T = \sum_{i=7}^{11} F_{x-i}(t)$$

$$\bar{T} \cong 0{,}00172\ N$$



5g. Cost of Transport



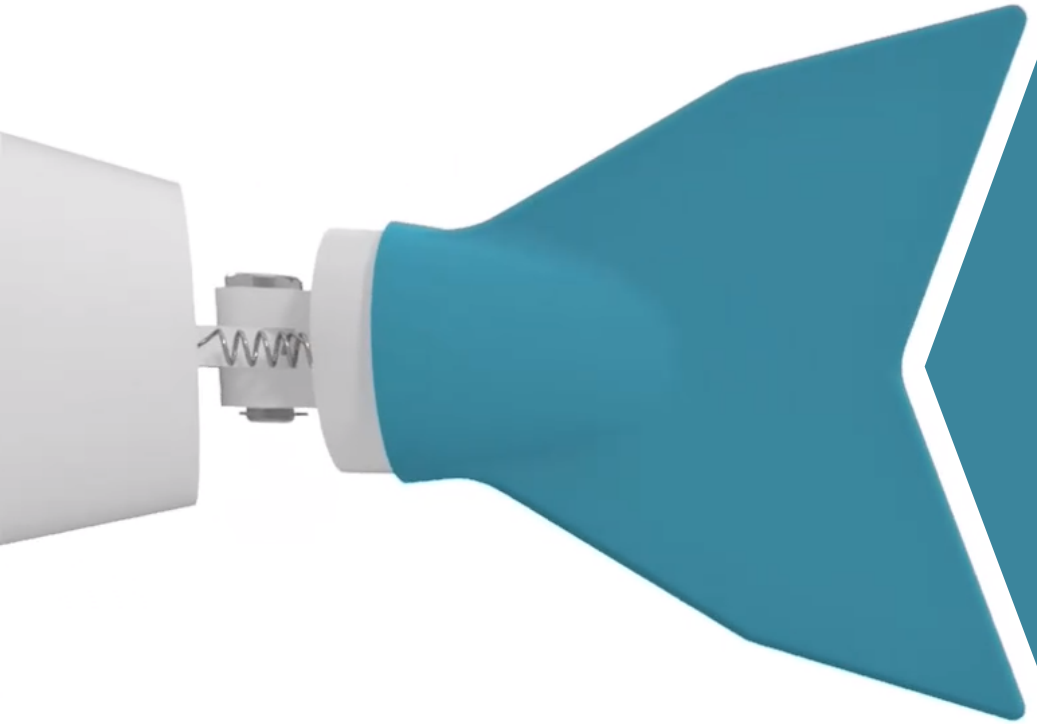
→ The CoT represents the ***energy expenditure*** to move *one unit of mass for one length units*. Can be shown how it changes as function of the ***frequency*** and ***amplitude***.

$$CoT = \frac{E}{m \cdot d} \left[\frac{J}{kg\ m}\right]$$

6. REFERENCES

[1] Sæþór Ásgeirsson. *Motion Analysis of Arctic char and Atlantic salmon*. Motion Analysis of Arctic char and Atlantic salmon. *2013*

[2] Wisdom E. K. Agbeti; Arjan P. Palstra; Suzy Black; Leonardo Magnoni; Martin Lankheet; Hans Komen. *Swimming at Increasing Speeds in Steady and Unsteady Flows of Atlantic Salmon Salmo salar: Oxygen Consumption, Locomotory Behaviour and Overall Dynamic Body Acceleration*. Biology. *2024*



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